NAUD TAFESSE

Email: ndtafesse@gmail.com **Address:** Springfield, VA **Phone Number:** (703) 755-5849 **Website:** https://naudt03.github.io/

TECHNICAL SKILLS

Languages: JavaScript, Python, Java, HTML/CSS, C, Kotlin
Databases/Cloud: MongoDB, MySQL, AWS (ECS, EC2, RDS, Route 53, CloudFront)
Frameworks/Libraries: React, Material UI, Node.js, Express.js, Spring Boot, JPA/Hibernate, Jakarta EE, JUnit
Tools/Concepts: Git/GitHub, Gradle, Maven, Docker, JBoss, REST APIs, load balancing, SSL/TLS, data visualization

EXPERIENCE

Regal Abyssinia | *JavaScript, MongoDB, React, Spring Boot, Docker, AWS ECS* **Falls Church, Virginia**
Full-Stack Developer *May 2025 — August 2025*

- Designed and deployed a full-stack restaurant platform using React, Spring Boot, and MongoDB for online ordering, menu management, and user reviews.
- Built RESTful APIs and deployed the containerized application to AWS ECS Fargate with load balancing and HTTPS (SSL/TLS) for scalable, secure transactions.
- Implemented JWT-based authentication to protect user accounts and transaction flows.
- Improved platform performance and user experience, contributing to consistent 5-star customer feedback and a ~30% increase in online order revenue.

CodeKids VT | *JavaScript, HTML, CSS, React, Node.js* **Blacksburg, Virginia**
Undergraduate Researcher *January 2025 — May 2025*

- Developed interactive web-based educational content to teach AI, coding, and cybersecurity concepts to K-12 students, improving engagement and comprehension in STEM activities.

MachWorks | *C, Embedded Systems, Sensor Integration* **Blacksburg, Virginia**
Avionics Developer *May 2024 — December 2024*

- Designed and implemented a real-time sensor integration system in C to monitor mini-plane altitude with sub-meter accuracy.
- Engineered low-latency height readings that improved flight stability, navigation reliability, and hardware/software responsiveness during test operations.

RELEVANT PROJECTS

AI Cologne Identification System | *Python, Flask, scikit-learn, NumPy, Pandas, Raspberry Pi, BME688*

- Built an edge-based machine learning system that identifies cologne profiles from VOC signatures captured by a Bosch BME688 gas sensor.
- Engineered feature extraction and classification pipelines, including heater-step fingerprinting and RandomForest-based real-time prediction on Raspberry Pi hardware.
- Developed a live Flask dashboard for on-device inference and sensor visualization, with safeguards for abnormal readings and hardware instability.

AI Crisis Events Web Crawler | *Flask, React, scikit-learn, MongoDB, Docker*

- Developed a full-stack web application for training models, configuring crawl parameters, and analyzing crisis-related content through dynamic graphs and URL lists.
- Integrated keyword-based and one-class classification workflows into an interactive frontend/backend pipeline.
- Visualized crawl data using vis-network.js and containerized the application with Docker for consistent deployment.

Vacation Plan Assistant | *Jakarta EE, PrimeFaces, MySQL, AWS EC2, Gradle, Wildfly (JBoss)*

- Developed a cloud-based travel planning application with flight comparison, vacation reviews, weather forecasting, and account-based trip tracking.
- Implemented MySQL-backed persistence with JPA/Hibernate and integrated Amadeus, Open-Meteo, and Google Maps APIs for real-time travel, weather, and location data.

Custom Linux OS Shell | *C, Unix*

- Built a C-based Linux shell supporting commands such as cd, fg, bg, and history, along with piping and terminal job control.
- Implemented foreground/background job handling and input/output redirection for multi-process command execution.

EDUCATION

Virginia Polytechnic Institute and State University **Blacksburg, Virginia**
Bachelor of Science in Computer Science *August 2021 — May 2025*
Relevant Coursework: Artifical Intelligence in Python, Software Design & Data Structures, Computer Organization, Data Structures & Algorithms, Computer Systems, GUI Programming/Graphics, Cloud Software Development, Mobile Software Development, Data & Algorithm Analysis